



DOW™ Electrical & Telecommunications HFDB-4201 NT Crosslinkable Power Cable Insulation Compound

Overview

HFDB-4201 NT is an unfilled, crosslinkable, low density polyethylene insulation compound designed for medium voltage power cable insulation applications. HFDB-4201 NT has been designed to have a low level of additive bloom for a long storage life, low dusting and an enhanced degree of scorch retardance.

HFDB-4201 NT is recommended for the insulation of power distribution cables.

Specifications:

HFDB-4201 NT is designed for use in power distribution cables. Cables insulated with HFDB-4201 NT, using sound commercial manufacturing practice, would be expected to meet the latest editions of the following specifications and regulations:

- ANSI/ICEA*: S-94-649, S-97-682, S-93-639/NEMA WC74
- AEIC* CS8
- UL 1072
- CENELEC HD 620 S2
- IEC 60502
- GB/T 12706
- BSI BS 6622, 7870-4

* Applicable to USA manufactured product only.

Physical	Nominal Value (English)	Nominal Value (SI)	Test Method
Density ¹	0.920 g/cm ³	0.920 g/cm ³	ASTM D792
Mechanical	Nominal Value (English)	Nominal Value (SI)	Test Method
Tensile Strength	2900 psi	20.0 MPa	ASTM D638
Tensile Elongation (Break)	500 %	500 %	ASTM D638
Thermal	Nominal Value (English)	Nominal Value (SI)	Test Method
Hot Set			IEC 60811-2-1
Under Load : 392°F (200°C)	< 100 %	< 100 %	
Without Load : 392°F (200°C)	< 10 %	< 10 %	
Aging	Nominal Value (English)	Nominal Value (SI)	Test Method
Change in Tensile Properties - 7 days @ 275°F/135°C	%	%	ASTM D638
Electrical	Nominal Value (English)	Nominal Value (SI)	Test Method
Volume Resistivity (73°F (23°C))	> 1.0E+15 ohms·cm	> 1.0E+15 ohms·cm	ASTM D257
Dielectric Strength			ASTM D149
0.125 in (3.18 mm), Method A (Short-Time)	760 V/mil	30 kV/mm	
0.125 in (3.18 mm), Method B (Step-by-Step)	580 V/mil	23 kV/mm	
0.125 in (3.18 mm), Method C (Slow Rate-of-Rise)	> 990 V/mil	> 39 kV/mm	
Dielectric Constant (73°F (23°C))	2.30	2.30	ASTM D150
Dissipation Factor (73°F (23°C))	3.0E-4	3.0E-4	ASTM D150

Additional Information

Cleanliness Requirements:

HFDB-4201 NT meet standards for cleanliness established for an unfilled, crosslinkable distribution cable insulation compound. Throughout the production process, the product is tested to ensure a specific level of cleanliness. Extruded tapes are scanned by an automatic inspection system in a clean room. The purity data is managed using an acceptance sampling procedure, which ensures that the product meets or exceeds the referenced standards.

Processing Techniques:

HFDB-4201 NT provides excellent performance and outstanding output rates over a range of extrusion conditions. For optimum results, melt extrusion temperatures in the range of 115°C to 140°C (240°F to 285°F) are recommended, although higher melt temperatures may be possible on certain extrusion lines with due care. In general the use of a minimum 60 mesh screen pack system is recommended. However, specific processing recommendations can only be made when information about the application and actual extrusion and processing equipment types are known.

Storage:

The environment or conditions of storage greatly influences the recommended storage time. Storage under extreme conditions may affect the quality, processing, or performance of the product. Storage should be in accordance with good manufacturing practices. The recommended storage conditions, in the original unopened packages, are dry conditions with temperatures between 50°F and 86°F (10°C and 30°C). When stored between 50°F and 86°F (10°C and 30°C), the product may be used for up to one year from the date of sale or two years from the date of manufacture, whichever comes first. The recommended maximum storage time is 1 year at 104°F (40°C). It is recommended that the practice of using the product on a first-in / first-out basis be established

Packaging:

HFDB-4201 NT can be delivered in different packaging types dependent on the specific materials handling needs. These packaging types could be in top and bottom unloading UNICLEAN™ octabins as well as top and bottom unloading octabins. Please consult with your local Dow sales representative to discuss your packaging needs.

Extrusion	Nominal Value (English)	Nominal Value (SI)
Melt Temperature	240 to 280 °F	116 to 138 °C

Extrusion Notes

HFDB-4201 NT provides excellent surface finish and outstanding output rates over a broad range of conditions. For optimum results, melt extrusion temperatures in the range of 240 to 280°F (116-140°C) are recommended, although higher melt temperatures are possible on certain equipment with due care. Generally, a 60-40-20 screen pack is recommended. However, specific recommendations for processing conditions can be determined when the application and type of processing equipment are known.

Notes

These are typical properties only and are not to be construed as specifications. Users should confirm results by their own tests.

¹ 23°C

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This document is intended for use within Europe

Published: 2007-02-19

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